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Department of Computer Science

Embedded Operating System

HW01: Environment Setup & First Kernel Module

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Wed. 13:20 – 16:20 EDB06

Purpose for this Homework

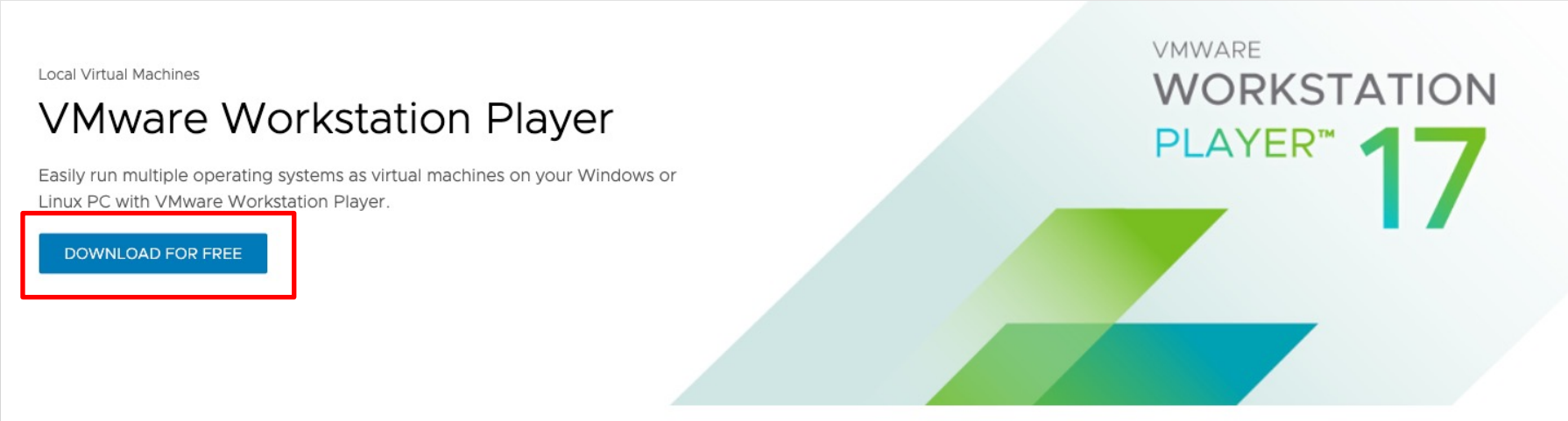
- **Part 1. Prepare Environment for future homework**
  - Install VMware
  - Download Ubuntu's iso
  - Install it on VMware's virtual machine
  - Screenshot your Ubuntu Desktop in your report
- **Part 2. Learn how to write a kernel module**
  - install build-essential package
  - new a folder
  - create hello_world.c & makefile in that folder
  - compile your code with `make` command
  - Screenshot the results in your report
    - Your hello_world.c & Makefile
    - Check your module after insmod with lsmod
    - demsg

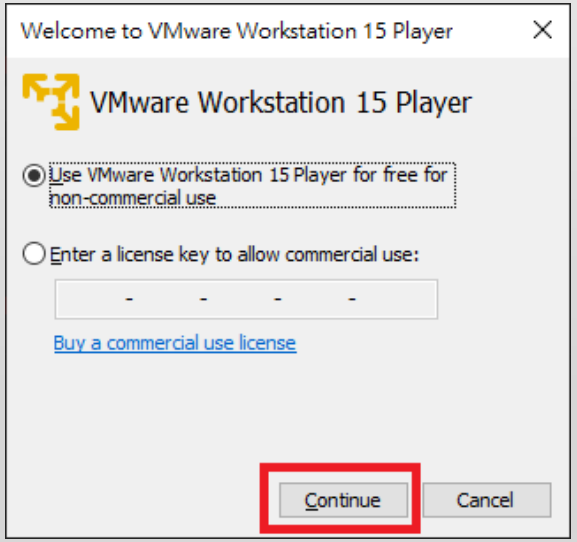
Purpose for this Homework

- **Part 3. Learn how to read files in a kernel module**
  - New a folder
  - Create fileIO.c & Makefile in that folder
  - Change Makefile for new module name
  - The file is input.txt and you have to save the content to output.txt in a reversed order
  - Screenshot your output.txt in your report

Install VMware Workstation Player

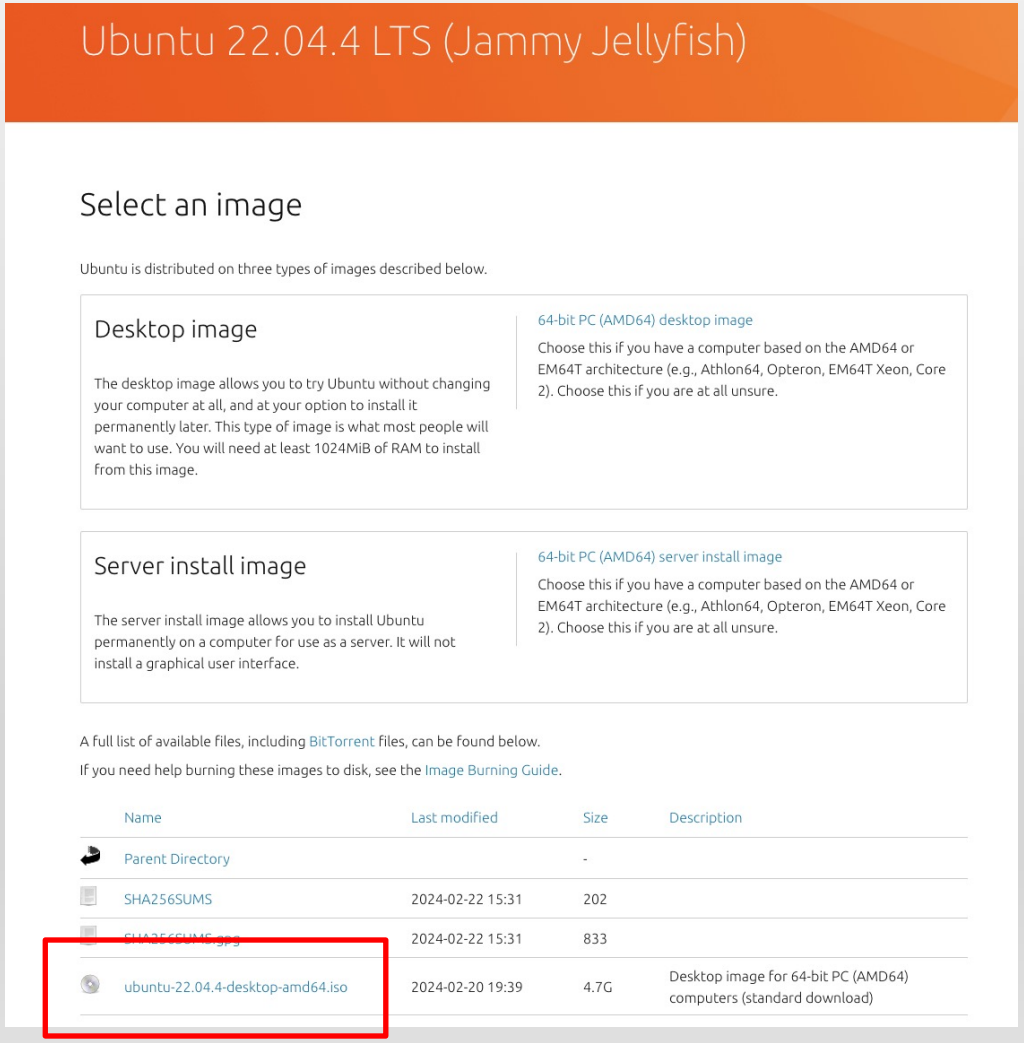
- Go to [VMware Workstation Player | VMware]
  - Download the VMware setup file
  - Execute it and keep clicking “Next” to install it





Download Ubuntu's iso

- Go to [Ubuntu 22.04.4 LTS (Jammy Jellyfish)]



Ubuntu 22.04.4 LTS (Jammy Jellyfish)

Select an image

Ubuntu is distributed on three types of images described below.

Desktop image

The desktop image allows you to try Ubuntu without changing your computer at all, and at your option to install it permanently later. This type of image is what most people will want to use. You will need at least 1024MiB of RAM to install from this image.

64-bit PC (AMD64) desktop image

Choose this if you have a computer based on the AMD64 or EM64T architecture (e.g., Athlon64, Opteron, EM64T Xeon, Core 2). Choose this if you are at all unsure.

Server install image

The server install image allows you to install Ubuntu permanently on a computer for use as a server. It will not install a graphical user interface.

64-bit PC (AMD64) server install image

Choose this if you have a computer based on the AMD64 or EM64T architecture (e.g., Athlon64, Opteron, EM64T Xeon, Core 2). Choose this if you are at all unsure.

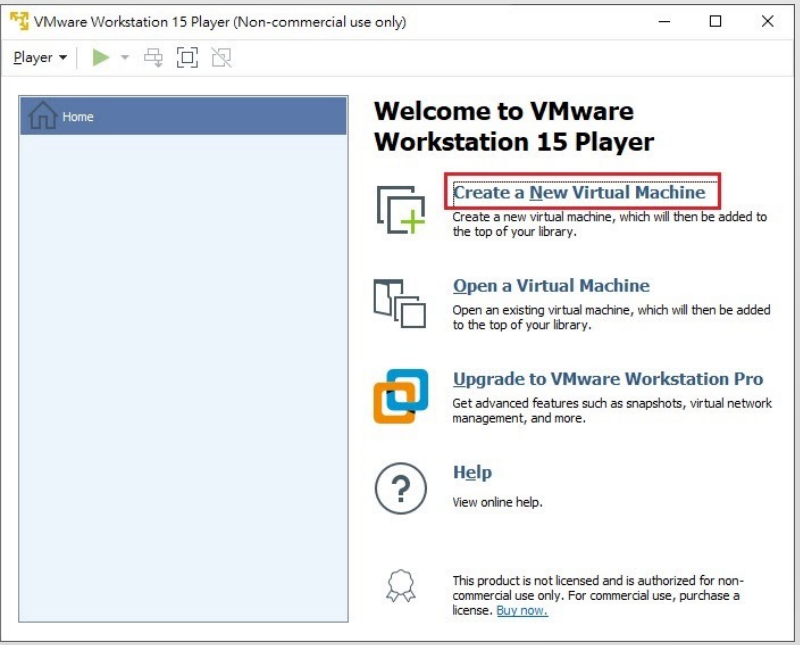
A full list of available files, including BitTorrent files, can be found below.

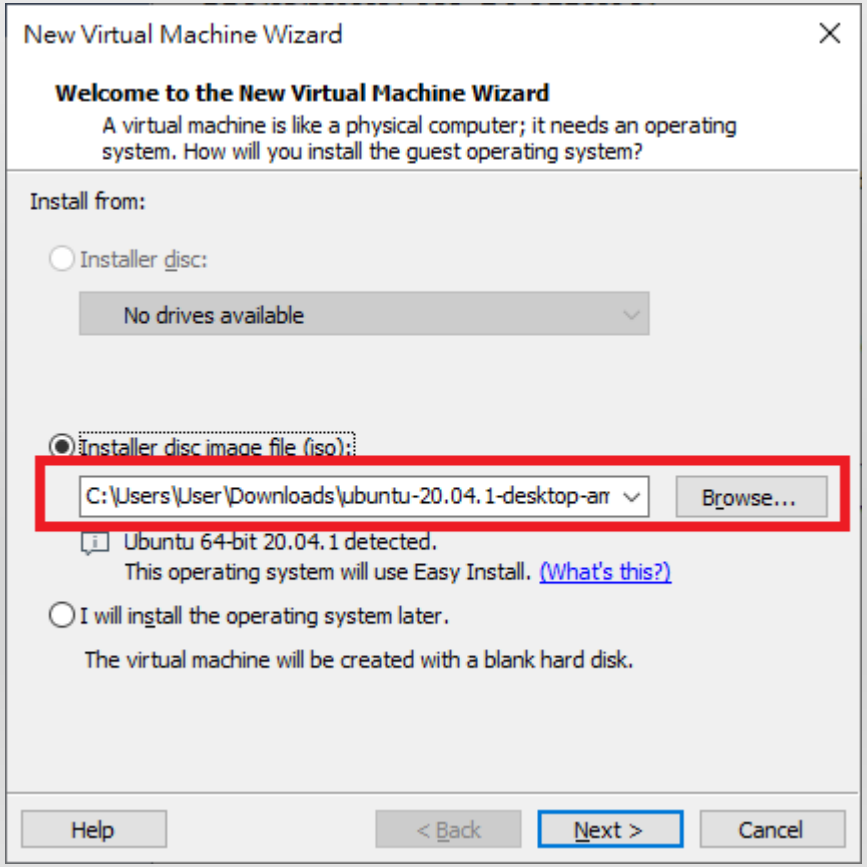
If you need help burning these images to disk, see the Image Burning Guide.

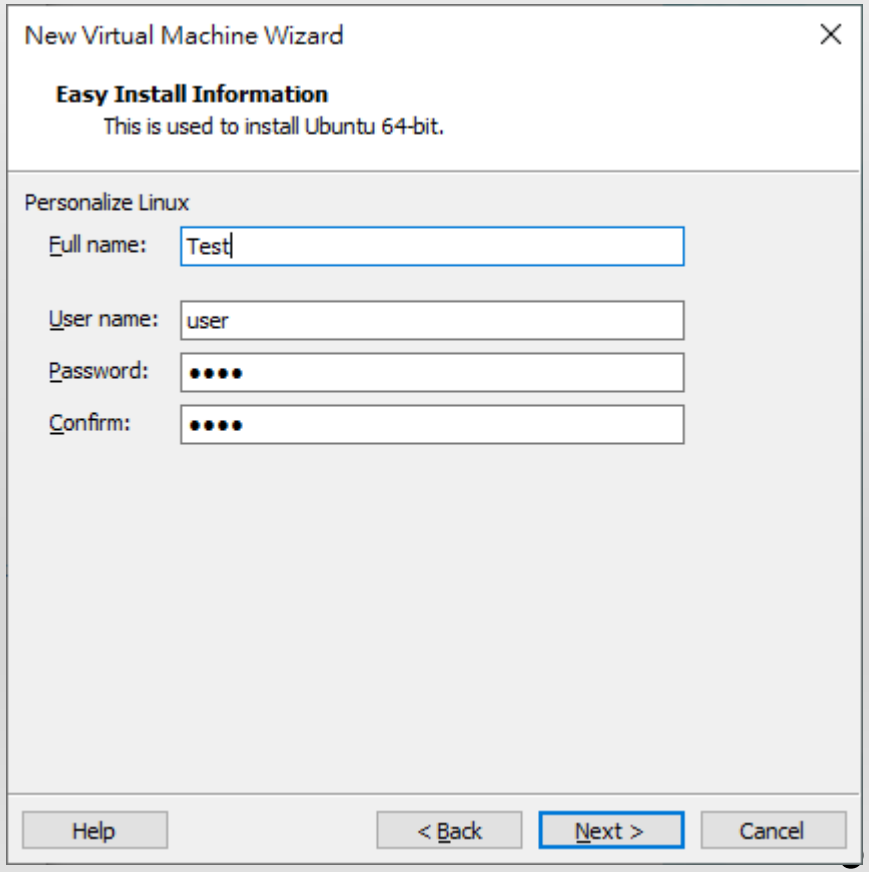
Name	Last modified	Size	Description
Parent Directory		-	
SHA256SUMS	2024-02-22 15:31	202	
SHA256SUMS.gpg	2024-02-22 15:31	833	
ubuntu-22.04.4-desktop-amd64.iso	2024-02-20 19:39	4.7G	Desktop image for 64-bit PC (AMD64) computers (standard download)

Install Ubuntu on Virtual Machine

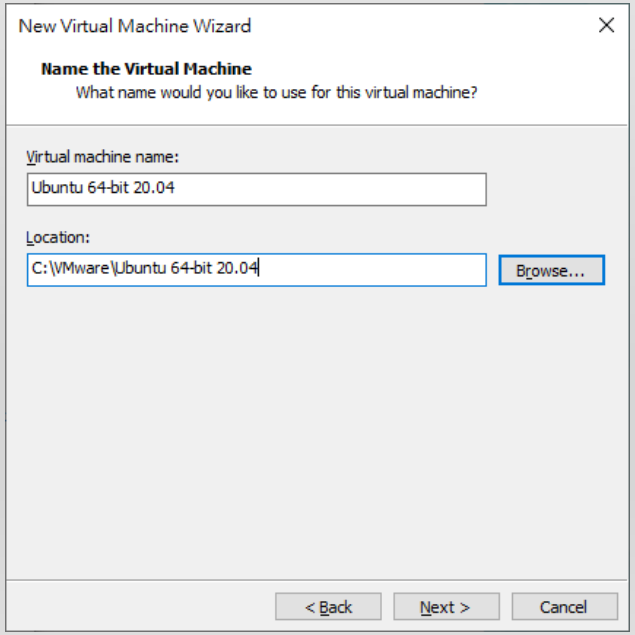
- Click “Create a New Virtual Machine”
- Choose the iso file you just downloaded
- Type the user name and password you like

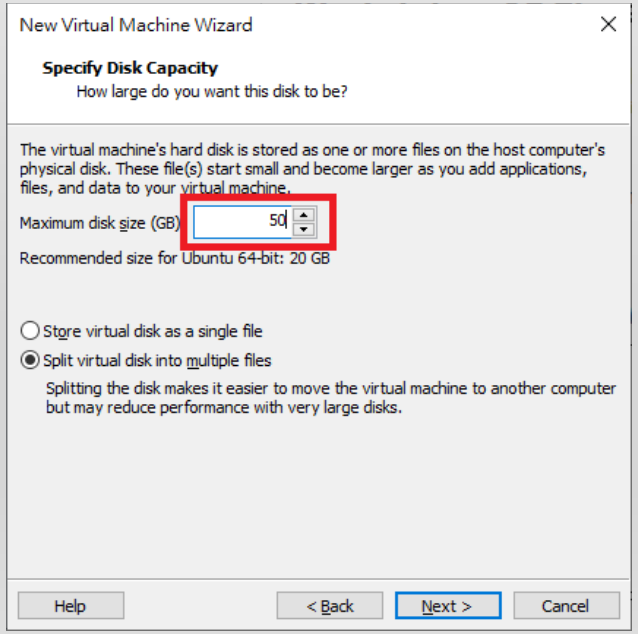


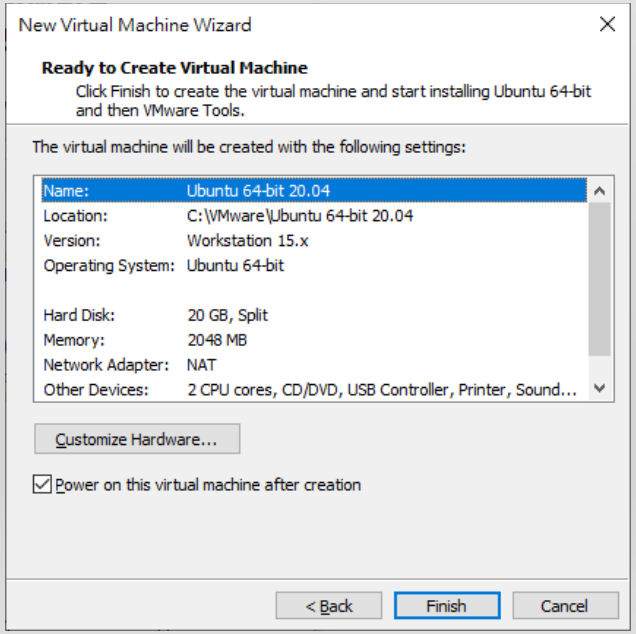


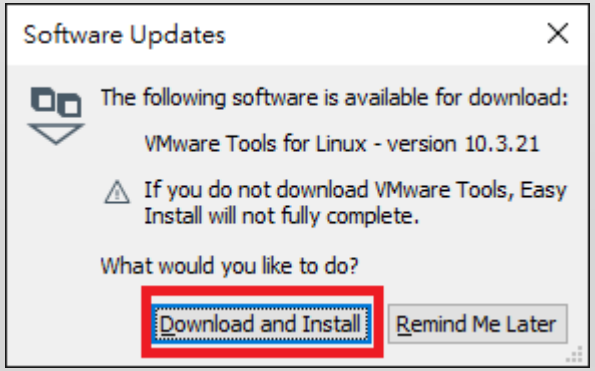


- Click “Next”.(or you may change the location of your virtual machine here)
- Set the disk size to at least 50GB
- Click “Next”
- Click “Download and Install”
- Wait for the virtual machine for Installing Ubuntu



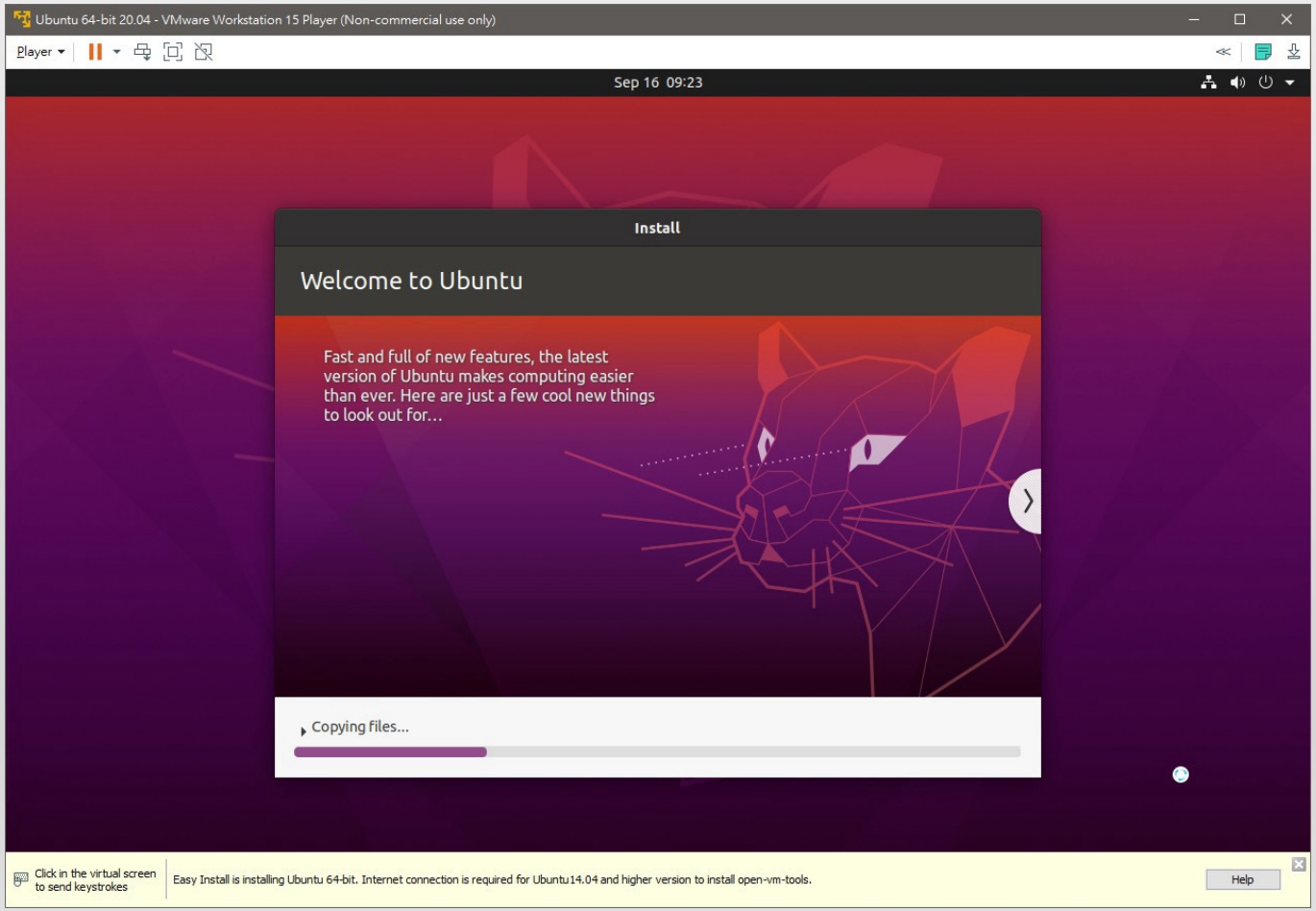






Screenshot desktop of your virtual machine

- Login into your virtual machine
- Screen shot the desktop and put it into report



Writing the Hello World Kernel Module

(In terminal / through UI)

1. sudo apt-get install build-essential
2. sudo apt-get install vim
3. mkdir hello_world
4. Go to the folder and create two files
   1. hello_world.c
   2. Makefile
5. Writing the Hello World kernel Module
6. Writing the makefile
7. vim Makefile, then press tab
   - Check if the format of your make file is correct
   - Incorrect will be marked with be marked in red

Your Hello World will look like this

```c
// kernel_world/hello_world.c

#include <linux/init.h>
#include <linux/module.h>

MODULE_DESCRIPTION("hello_world");
MODULE_LICENSE("GPL");

static int hello_init(void)
{
        printk(KERN_INFO "Hello world !\n");
        return 0;
}

static void hello_exit(void)
{
        printk(KERN_INFO "Bye !\n");
}

module_init(hello_init);
module_exit(hello_exit);
```

Your Makefile will look like this

```
PWD := $(shell pwd)
KVERSION := $(shell uname -r)
KERNEL_DIR = /usr/src/linux-headers-$(KVERSION)/

MODULE_NAME = hello_world
obj-m := $(MODULE_NAME).o

all:
        make -C $(KERNEL_DIR) M=$(PWD) modules
clean:
        make -C $(KERNEL_DIR) M=$(PWD) clean

```

After make, try the build module

1. If you encounter bug “gcc-12 not found”, type following command –> sudo apt install gcc-12 libgcc-12-dev
2. Mount the module -> sudo insmod hello_world.ko
3. Check if it’s mounted -> sudo lsmod | grep "hello_world“

```
qoolili@ubuntu:~/hello_world$ sudo insmod hello_world.ko
qoolili@ubuntu:~/hello_world$ sudo lsmod | grep "hello_world"
hello_world            16384  0
```

4. Unmount the module -> sudo rmmod hello_world
5. Go to dmesg to find the print message -> dmesg

```
[ 1675.554896] Hello world !
[ 1729.246413] Bye !
```

Please write Part 3 on your own

- You can do google, but please write your code by yourself
- Evaluation Method:
  - Download your input.txt, fileIO.c, and Makefile files.
  - Use your Makefile to compile your kernel module.
  - Run insmod to insert the module and rmmod to remove the module.
  - Validate that the contents of output.txt are the reverse of input.txt.
- Hints:
  - Do not use absolute paths in your code.
  - Create the output.txt file within your kernel module implementation.

Submission

- Submit your PDF report and source code in a zip format to the google Form.
- The google form will be published in Teams before the Deadline.
- The zip folder structure should look like this:
  - HW01_studentID.zip
    - report.pdf
    - hello_world
      - hello_world.c
      - Makefile
    - file_io
      - fileIO.c
      - Makefile
      - input.txt

Grading

- Part1 Ubuntu installation - 5%
- Part2 Hello World module - 25%
- Part3 File IO module - 70%

Q & A

Remember to submit your report.

The deadline is set as 3/27 23:59

Late submission will be not accepted!